

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2023

Course: B. Tech.

Branch: Civil Engineering

Semester: IV

Subject Code & Name: Structural Mechanics -I (BTCVC 403)

Max Marks: 60

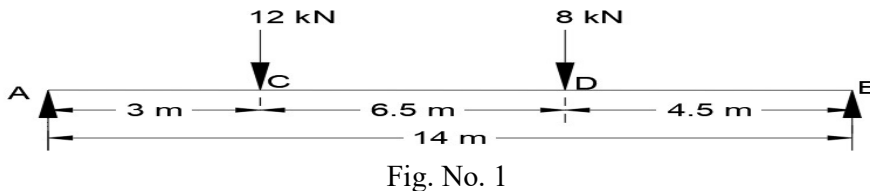
Date: 20/01/2024

Duration: 3 Hrs.

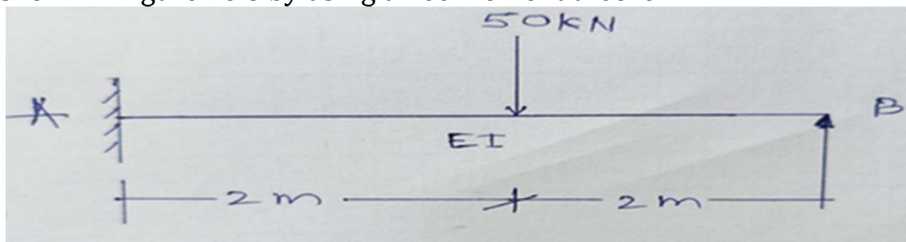
Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q.1 Solve Any Two of the following.		12
A) Derive the differential equation for Slope and Deflection.	CO-1	6
B) A wooden beam 140 mm wide and 240 mm deep has a span of 4 m. Determine the load, that can be placed at center to cause deflection of 10 mm. Take E as 6 GPa.	CO-1	6
C) A Horizontal girder having uniform cross section is 14 m long and is simply supported at its end. It carries two concentrated loads as shown in fig. no. 1 Calculate the deflection of the beams under the loads C and D. Take E = 200 GPa and $I = 160 \times 10^6 \text{ mm}^4$.	CO-1	6



Q.2 Solve Any Two of the following.		12
A) State and explain the Maxwell's reciprocal Theorem.	CO-2	6
B) Using energy theorem, find the deflection at free end of cantilever beam of span l carries a point load at the free end and mid span.	CO-2	6
C) A rolled steel joist ISMB 250 x 125 mm as shown in fig. carries a single concentrated load of 20 kN at the right third point over a simply supported span of 9 m. if the values of I_{xx} for the beam is $51.316 \times 10^6 \text{ mm}^4$ and the values of E for the material is 200 GPa, Calculate by the use of conjugate beam method Deflection under the load	CO-2	6
Q.3 Solve Any Two of the following.		12
A) Define determinacy and indeterminacy of structures. Explain in detail with examples	CO-3	6
B) Calculate the support moment and draw BMD and SFD for a beam as shown in figure no 6 by using three moment theorem.	CO-3	6



- C) A fixed beam of span 'L' subjected to an external clockwise moment 'M' at a distance 'a' from one of its fixed supports and at a distance 'b' from other end fixed support. Determine the end moments developed.

CO-3 6

Q.4 Solve Any Two of the following.

12

- A) Explain the following terms

CO-4 6

- i) Carry Over Factor
- ii) Distribution factor
- iii) Stiffness Factor.

- B) Analyze the frame shown fig. no. 3 for moments at the ends of members. Draw BMD. EI is same for all the members. Use Moment distribution method.

CO-4 6

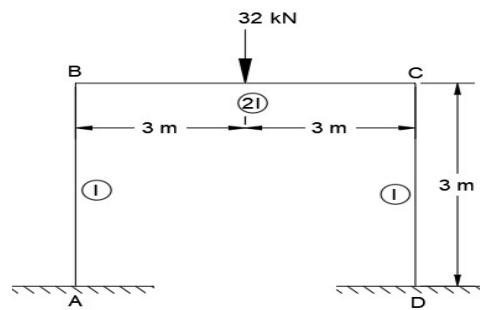


Fig. No. 3

- C) Calculate support moments by moment distribution method for given continuous as shown in fig.

CO-4 6

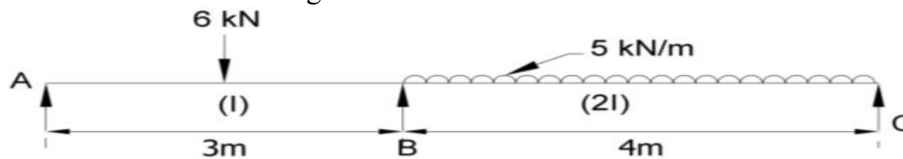


Fig. No. 4

Q.5 Solve Any Two of the following.

12

- A) State the assumptions made in slope and deflection method for determining the support moments.
- B) A continuous beam ABC is fixed at A and simply supported at B and C. The Span AB of 6 m and carries a UDL of 100 kN/m. The span BC is 4m carries a UDL of 300 kN/m . Determine the support moments.
- C) Explain the procedure for analyzing a statical indeterminate structure using slope and deflection method.

CO-5 6

CO-5 6

CO-5 6

*** End ***